Fiscal Year:	FY 2019	**Task Last Updated:**	FY 12/07/2021
PI Name:	Juran, Cassandra Ph.D.		
Project Title:	The Regulatory Effect of Gravity Loading on p21 Expression in Mouse Stem Cells: Proliferation, Osteoblast Differentiation, and Osteoclastogenesis Control		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	cassandra.m.juran@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	954-275-0136
Organization Name:	NASA Ames Research Center		
PI Address 1:	Blue Marble Space Institute of Science Affiliate Research Scientist (BMSIS)		
PI Address 2:	MS 236-7		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	03/02/2016	**End Date:**	03/01/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/1/2019 per PI (Ed., 8/9/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo Ph.D. (MENTOR/ NASA Ames Research Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Postdoctoral Program Fellow Long-duration exposure to space has deleterious effects on the human body including loss of significant bone mass due in part to improper balance of bone synthesis and bone resorption cells. Observation of mice experiments conducted on the International Space Station (ISS) have demonstrated that a stem cell senescence modulator protein CDKN1a/p21 is upregulated in microgravity. This proposal will utilize a two experiment research approach to 1) evaluate how hypergravity culture influences p21 expression and 2) investigate the regulatory effects of p21 expression on mechanosensory signaling, specifically the wnt-signaling cascade a key osteoinductive mechanotransductive pathway. Expected results will enable further research into the mechanisms of microgravity induced bone mass loss through stem cell inactivation and supply early evidence of a potential strategy for recovery of osteogenesis after spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis and other aging related bone degeneration and disuse diseases are presently remediated by symptom minimizers rather than treated at the source of the problem. Spaceflight is an aging homologue, making our fundamental space bone biology research applicable to clinical investigations done on ground. In addition to the obvious clinical applications of our work, many novel technological breakthroughs are instigated through our basic biology work including flight protocol for novel end-to-end sample to sequence molecular workflows applicable to spaceflight and single cell “quantum” genomics providing an individual cell resolution of gene expression.
Task Progress:	(Ed. note December 2021: Final report received for Dr. Juran's NPP, dated June 2019.) Since starting my NASA Postdoctoral Program (NPP) fellowship I have made significant progress in experimentally addressing all of my aims. I made significant discoveries on the mechanistic aspect of p21 regulation of gravity mechanotransduction and established experimental procedures critical for continued investigation of the cell based molecular mechanism cascades responsible for bone mechanotransduction. Tasks accomplished during my first year included re-initializing and modernizing many of the mechanical stimulator cell culture bioreactor systems including a mechanical stretch, 3D fluid shear, and hypergravity centrifugal cell culture system. Using model cell lines MLOY4 osteocyte-like cells and MC3T3-E1 osteoblast-like cells I investigated mechanosignaling cascades involved in homeostasis maintenance, stem cell recruitment, and direct cell-cell communication. Mechanotransduction effects of various 2D and 3D mechanical regimes on bone cells have provided foundational data on the mechanisms of translation of mechanical signaling to molecular regulation of bone phenotype and structure. These results have indicated that key regulatory pathways like gap junctional communication and wnt-signaling are significantly modulated by the type of load and the intensity of the force imparted. That data allowed me to test the regulatory role of CDKN1a/p21 in bone marrow stem cell mechanoregulation using a custom designed cyclic stretch mechanostimulation osteogenesis bioreactor for culturing and maturing osteoblast cells from bone marrow stem cells. The results of the stretch mechanical stimulation studies have elucidated the interesting regulatory effect of CDKN1a/p21 on progenitor proliferation and differentiation. Our critical finding and developing hypothesis at the conclusion of year 3 of my NPP project is: CDKN1a/p21 expression is suppressed by mechanical stimulation through the action of gravity dependent mechanotransductive pathways that positively promote cellular regeneration. This working hypothesis will continue to be developed as I transition to a research position at NASA Ames. The regulatory effect of CDKN1a/p21 on specific mechanosignaling pathways is still under investigation and will be tested in true microgravity during the upcoming Rodent Research 10 spaceflight mission, for which I am a Co-Investigator. General conclusions: Mechanotransduction in the bone is complex, with no single cell type responsible for translation of mechanical stress into molecular regulation. Osteocyte cells have more sensitivity to perturbation in mechanical load while osteoblasts respond to static loading, like gravity loading indicating that for maintenance of bone both dynamic and static loading is critical. CDKN1a/p21 is a mechanoregulator of proliferation and differentiation. Osteoblastogenesis culture of CDKN1a/p21 knockout cells under cyclic stretch increases cell turnover and differentiation phenotype expression resulting in highly mineralized dense cell networks. A Future assessment of CDKN1a/p21’s mechanoregulatory ability over osteogenesis will evaluate how CKDN1a/p21 sensitizes aging osteoprogenitor cell population, to test if p21 plays a role in stem cell exhaustion seen as a function of aging. Also the Bone and Cell Signaling team has plans to use novel single cell expressome (10X MinIon) techniques are planned for the future to resolve individual cell responses in mixed primary cell populations and obtain greater insight into gravity mechanotransduction mechanisms. Publications and Accepted and Presented Conference Abstracts Juran, Cassandra M., et al. "Cdkn1a deletion or suppression by cyclic stretch enhance the osteogenic potential of bone marrow mesenchymal stem cell-derived cultures." Stem cell research 56 (2021): 102513. Moore, Marc C., Van De Walle, A., Chang, J., Juran, CM., and Peter S. McFetridge. "Human Perinatal-Derived Biomaterials." Advanced Healthcare Materials 2017: 6(18): 1700345. Juran, C.M., Zvirblyte, J., and Almeida, E.A.C. The Role of Gravity Mechanotransduction in Regulating Stem Cell Tissue Regenerative Potential at the Single Cell Expressome Level. International Symposium Stem Cells & Organoids in Development & Disease, 2019 Amsterdam, Netherlands. Juran, CM, Blaber, E, and Almeida, EA. Modulation of Bone Marrow Primary Cell Osteoblastogenesis and Cell Senescence by Mechanical Stimulation. American Society for Gravitational and Space Research (ASGSR) October 2017, Seattle WA. Ma, K, Juran, CM, and Almeida, EA. Novel Approach to Quantification of Telomere Length with Direct Nanopore Sequencing and PCR Amplification. American Society for Gravitational and Space Research (ASGSR) October 2017,

	Seattle WA. Juran, CM, Blaber, E, and Almeida, EA. Effects of modified gravity coupled with mechanical stimulation on molecular signal transduction and target gene transcription in 3D osteon cell network. International Astronautical Congress September 2017, Adelaide, Australia. Juran, CM, Blaber, E, and Almeida, EA. Exaggerated Mechanical Strain Modulates Functional Outcome of MSC Osteoblastogenesis. NextGen Stem Cell August 2017, Hartford, CT. Juran, CM, Smith, DJ, and Almeida, EA. Novel End-to-End Molecular Biology Approach to Direct Nanopore 1D Sequencing on ISS. ISS R&D June 2017, Washington DC, USA. Juran, CM, Blaber, E, Almeida, EA. Osteogenic signaling control of MSCs regulated by exaggerated mechanical loading of osteocyte-osteoblast cellular network. Cellular and Molecular Bioengineering (CMBE) January 2017, Kohala Coast, The Big Island of Hawaii, HI. Juran, CM, Blaber, E, Almeida, EA. Mechanotransductive regulation of gap-junction activity between MLO-Y4 osteocyte-like and MC3T3-E1 osteoblast-like cells in Three-Dimensional Co-Culture. American Society for Gravitational and Space Research (ASGSR) October 2016, Cleveland, OH.
Bibliography Type:	Description: (Last Updated: 04/22/2024)
Articles in Peer-reviewed Journals	Juran CM, Zvirblyte J, Cheng-Campbell M, Blaber EA, Almeida EAC. "Cdkn1a deletion or suppression by cyclic stretch enhance the osteogenic potential of bone marrow mesenchymal stem cell-derived cultures." Stem Cell Res. 2021 Oct;56:102513. Epub 2021 Aug 26. https://doi.org/10.1016/j.scr.2021.102513 ; PMID: 34517335 , Oct-2021